**Note**

When $k = 0$, by definition,

$$E(1) = \begin{cases} \sum_{x} f(x) = 1 & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} f(x)\,dx = 1 & \text{if } X \text{ is continuous} \end{cases}$$

## 11.5.2 Variance

Variance is a statistical measure that tells us how measured data vary from the average value of the set of data. Mathematically, variance is the mean of the squares of the deviations from the arithmetic mean of a data set. The terms variability, spread, and dispersion are synonyms, and refer to how spread out a distribution is.

**Definition 11.9: (Variance)**

*The **variance** of a random variable $X$* denoted by $\text{Var}(X)$ or $V(X)$ or $\sigma^2$ (or $\sigma_x^{\,2}$) is

$$V(X) = E(X - E(X))^2 = E(X - \mu)^2$$

Square root of variance is called standard deviation. That is standard deviation $\sigma = \sqrt{V(X)}$. The variance and standard deviation of a random variable are always non negative.

## 11.5.3 Properties of Mathematical expectation and variance

(i) $E(aX + b) = aE(X) + b$, where $a$ and $b$ are constants

**Proof**

Let $X$ be a discrete random variable

$$\begin{aligned} E(aX + b) &= \sum_{i=1}^{\infty} (ax_i + b) f(x_i) && \text{(by definition)} \\ &= \sum_{i=1}^{\infty} \left(ax_i f(x_i) + bf(x_i)\right) \\ &= a\sum_{i=1}^{\infty} x_i f(x_i) + b\sum_{i=1}^{\infty} f(x_i) \\ &= aE(X) + b(1) && \left(\because \sum_{i=1}^{\infty} f(x_i) = 1\right) \end{aligned}$$

$$E(aX + b) = aE(X) + b.$$ ■

Similarly, when $X$ is a continuous random variable, we can prove it, by replacing summation by integration.

| | | |
|---|---|---|
| Corollary 1: | $E(aX) = aE(X)$ | ( when $b = 0$ ) |
| Corollary 2: | $E(b) = b$ | (when $a = 0$ ) |



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